

K2-30b

R-2516

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_181118 · created 2026-08-25T00:47:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458440.8867 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.047
Transit depth (Rp/Rs)^2	1.7 +/- 1.24 [%]
Orbital Inclination (inc)	84.72 +/- 2.36
Airmass coefficient 1 (a1)	0.51 +/- 0.29
Airmass coefficient 2 (a2)	0.65 +/- 0.51
Scatter in the residuals of the lightcurve fit is	70.94 %
Best Comparison Star	#1 - [240, 121]
Optimal Aperture	2.16
Optimal Annulus	10.76
Transit Duration (day)	0.044 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.047 vs 0.1266 (deviation 0.09σ)
Duration fit vs published	0.044 d vs 0.0971 d (deviation 1.23σ)
Mid-time O–C	-12.6 min
Depth SNR	2.8
Residual scatter	70.94 %

Light curve

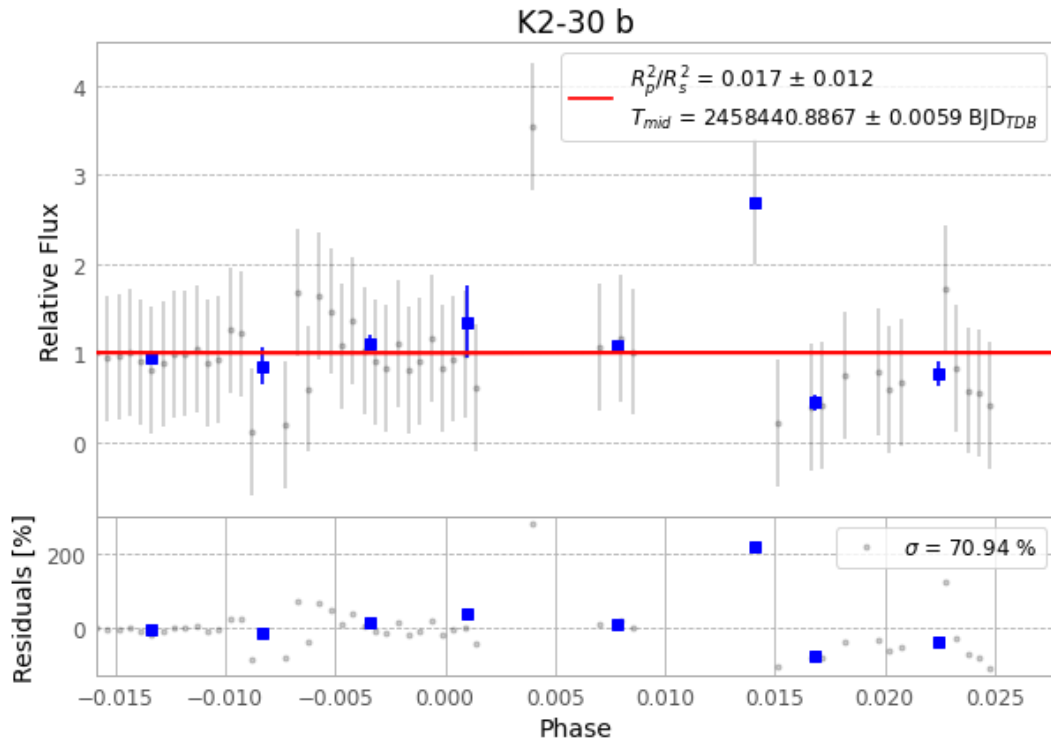


Figure 1 — FinalLightCurve_K2-30 b_2018-11-18.png (SHA-256 175cb44a89fbabd9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 312], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4bfdd44d0297c44f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), K2-30181118073312.FITS → K2-30181118115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-181118.log (40978e68427d...), FinalLightCurve_K2-30 b_2018-11-18.png (175cb44a89fb...), FinalLightCurve_K2-30 b_2018-11-18.csv (ec898ec5d7e6...)

Compute resources

Wall time 220.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:47Z.